

SORAYA REMAILI

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EDUCATION

University of South Carolina Honors College, Columbia, SC
B.S. in Computer Science, B.S. in Statistics, and B.S. in Biology with Honors
Expected May 2027
GPA: 4.0/4.0

Thomas Jefferson High School for Science and Technology, Alexandria, VA
High School Diploma, June 2023
GPA: 4.45 (weighted scale)
AP Scholar with Honor

The Paris Institute for Political Science, Paris, FR
Summer Certification, July 2022
Final Grade: 17.75/20
Focus on International Negotiation and Affairs

HONORS AND GRANTS

University of South Carolina President's List, 2023–Present
University of South Carolina Academic Excellence Scholar, \$2000/year
University of South Carolina Undergraduate Scholarship, \$25000/year
South Carolina Honors College Undergraduate Research Grant, 2024–2026, \$6000
Carter Bays Endowed Scholarship for Computer Science, 2024–2025, \$1000
Vlahoplus Engineering Honors Scholarship, 2025–2026, \$1000
Provost AI Undergraduate Fellowship, 2026, \$5000
Magellan Mini Research Grant, 2026, \$750
SEAMAMMS Runner-Up Best Undergraduate Long Talk, 2026

POSTERS AND PRESENTATIONS

International Sclerochronology Conference, St. John's, NL **May 2026**
Wanamaker, A.D., A. Walton, S. Griffin, B. Black, A. Hauser, L. Jarosinski, N.M. Whitney, C.C. Ummenhofer, **S.S. Remaili**, A. Biastoch, T. Martin, and D.L. Thatcher.
Geochemical Signatures in the Long-lived Marine Bivalve *Arctica islandica* Show Intense Warming in the Down East Coastal Region of the Gulf of Maine.

International Sclerochronology Conference, St. John's, NL **May 2026**
Thatcher, D.L., H. Sun, A.D. Wanamaker, N.M. Whitney, L. Jarosinski, J.A. Stewart, B. Williams, M. LaVigne, C.C. Ummenhofer, **S.S. Remaili**, A. Biastoch, and T. Martin.
Master Shell Chronology and Multi-Proxy Geochemical Records Illustrate Oceanographic Variability in the Mid-Atlantic Region (USA) Since 1800 CE.

International Sclerochronology Conference, St. John's, NL **May 2026**
Whitney, N.M., A.D. Wanamaker, D.L. Thatcher, B. Williams, C.C. Ummenhofer, **S.S. Remaili**, A. Biastoch, T. Martin, H. Sun, L. Jarosinski, J. Berg, A. Walton, M. LaVigne, J. Stewart, E. Powell, R. Mann, E. Hofman, J. Klinck, N. Sprague, M. Spencer, A.K. Watkins, B. Martin, A. Hauser, R. Copich, and B. DePhillips.
Utilizing a Network of *Arctica islandica* Records and Model Simulations to Investigate Past Regional Ocean Dynamics in the Rapidly Warming Western North Atlantic.

SEAMAMMS, Wilmington, NC **April 2026**
Oral Presentation
Remaili, S.S.*, A.M. Kreuser, A.A. Sefah-Twerefour, E. Meyer-Gutbrod, and C.R. Edwards.
Lessons from Real-Time and Archival Acoustic Detections of North Atlantic Right Whales in the

Southeast US across Three Types of Platforms.

Discover USC, Columbia, SC

April 2026

Remali, S.S.*, A.M. Kreuser, A.A. Sefah-Twerefour, E. Meyer-Gutbrod, and C.R. Edwards.
Lessons from Real-Time and Archival Acoustic Detections of North Atlantic Right Whales in the Southeast US across Three Types of Platforms.

American Geophysical Union, New Orleans, LA

December 2025

Remali, S.S.*, N.M. Whitney, B. Williams, C.C. Ummenhofer, A. Biastoch, T. Martin, A.D. Wanamaker, D.L. Thatcher, J. Berg, A. Hauser, H. Sun, A. Walton, M.J. Mette, E. Powell, R. Mann, E. Hoffman, and J. Klinck.
Explaining Variability in Shell-Based Oxygen Isotope Ratios in the Western North Atlantic Using a Proxy System Model.

Discover USC, Columbia, SC

April 2025

Remali, S.S.*, A.M. Kreuser, A.A. Sefah-Twerefour, K. Dreger, E. Meyer-Gutbrod, and C.R. Edwards.
Pilot Program for Glider-Based Passive Acoustic Monitoring of Right Whales in the Southeast US Calving Ground (2025 version).

North Atlantic Right Whale Consortium, Providence, RI

October 2024

Society for Marine Mammalogy, Perth, WA

November 2024

Meyer-Gutbrod, E., C.R. Edwards, A.M. Kreuser, A.A. Sefah-Twerefour, K. Dreger, and **S.S. Remali**.
Pilot Program for Glider-Based Passive Acoustic Monitoring of Right Whales in the Southeast US Calving Ground (2024 version).

RESEARCH
EXPERIENCE

Active Acoustic ID of Arctic Fauna

May 2026–Present

Fulbright Canada-Mitacs Internship at Memorial University of Newfoundland

Research Directors: Dr. Frédéric Cyr and Dr. Maxime Geoffroy

- Analyze WBAT broadband acoustic profiles to identify fish and zooplankton across the polar front.
- Use Echoview and Python to process acoustic data from 38 kHz and 333 kHz transducers.
- Compare acoustic size estimates with spectra from LISST, LOPC, and UVP instruments.
- Develop unsupervised classification methods to distinguish fish and zooplankton in profiles without UVP data.

Skills: Echoview, broadband acoustics, active acoustic data processing, unsupervised classification, CTD-Rosette/WBAT data.

Water and Bill Payment Typologies

February 2026–Present

USC School of the Earth, Ocean, and Environment

Research Director: Dr. David Fuente

- Modeled Nairobi water billing and payment behavior using customer records from 2012–2023.
- Built Gaussian mixture models to classify customers into interpretable payment typologies.
- Identified high-arrears and low-payment customer groups to support utility revenue strategies.

Skills: XGBoost regression trees, Gaussian mixture models, customer segmentation, feature engineering, Python ML libraries.

Paleoceanographic Modeling

June 2025–August 2025

NSF-REU at Shannon Point Marine Center

Research Director: Dr. Nina Whitney

- Used Python to model paleoceanographic data and analyze historical climate variability.
- Applied multi-taper spectral analysis to identify patterns in climate and oceanographic records.

- Presented research findings at scientific symposiums and strengthened scientific communication skills.
- Skills:** Climate modeling, Python, Pyleoclim package usage, climate data product wrangling, multi-taper spectral analysis.

- NOAA Real-Time Analysis Workshop**

May 2024
- Gained hands-on experience with real-time data analysis techniques for environmental monitoring.
 - Collaborated with NOAA scientists and peers to analyze marine ecosystem data through simulations and case studies.

- Bioacoustic Data Analysis**

March 2024–Present
- USC School of the Earth, Ocean, and Environment
Research Director: Dr. Erin Meyer-Gutbrod
- Analyze acoustic data to identify baleen whale presence using Python, Raven, R, and LFDACS.
 - Compare detection methods by accuracy to evaluate their usefulness for whale conservation research.
 - Process and classify real-time acoustic data while developing statistical analysis and visualization skills.
- Skills:** Low-Frequency Detection Classification Software, R, RStudio, Python, Raven, real-time analysis, acoustic data processing, UNIX/LINUX.

- Jellyfish Bioremediation**

August 2021–June 2023
- Thomas Jefferson High School for Science and Technology
Research Director: Dr. Shawn Stickler
- Maintained jellyfish tanks and monitored water quality to support consistent experimental conditions.
 - Contributed to experimental design and data collection for aquatic ecosystem restoration research.
- Skills:** Research paper writing, jellyfish handling/care, tank maintenance, water quality analysis, experimental design, lab equipment care, aquatic ecosystem management.

- COMMUNITY
EXPERIENCE

Service Chair/DEI Chair

Spring 2024/Fall 2024
- Theta Tau Professional Engineering Fraternity
- Co-led Theta Tau’s service and equity committees, aligning projects with organizational goals.
 - Managed budgets and resources for service projects, securing additional funding when needed.

- Technology Adventure Camp Assistant Instructor**

July 2024
- Fairfax County Public Schools Summer Programs
- Taught children foundational robotics concepts, coding, and sensor technology in a hands-on environment.
 - Designed and facilitated engaging activities to teach computer skills and programming languages.

- SCIENCE
OUTREACH

Anacortes & Oak Harbor Public Library Touch Tank

July 2025
- Designed and executed a marine habitat learning experience for 100+ children.

- Guardianes Del Mar STEM Program**

June 2025
- Helped facilitate an intertidal zone and microscopy workshop for 15 middle-school students.

- North Atlantic Right Whale Festival**

November 2024
- Designed and taught activities to 300+ visitors to teach the importance of bioacoustics.

- PROJECTS

Pathways: Campus Connection Made Easy
- Built a full-stack student platform integrating campus navigation, events, courses, calendars,

and career planning tools.

- Developed the system using React, Django REST Framework, Docker, Redis, Celery, and GitHub Actions for deployment and automation.

DegreeBetter Academic Management System

- Full, back-to-frontend development of a degree planning system using Java, FXML, and Git.
- Developed the backend system using Java to handle user data, course requirements, and degree tracking.

Personal Website and Portfolio

- Built and deployed a responsive portfolio website using HTML, CSS, and JavaScript.
- Used GitHub Pages for hosting and Git for version control.

Tiny RISC-V Operating System

- Developed a bare-metal RISC-V operating system using C, Assembly, Makefile, and QEMU.
- Implemented a program loader, in-memory filesystem, spinlock synchronization, and permissions model.

Formula 1 Data Analysis

- Analyzed Formula 1 racing data in R to explore constructor performance, race geography, and championship trends.
- Cleaned, joined, and visualized datasets to compare driver results, positions, and historical team outcomes.

SKILLS

French: Professional working proficiency; Virginia State Seal of Biliteracy Recipient

Algerian Darija: Limited working proficiency; completed courses at Johns Hopkins CTY

Technical: R/RStudio, Python, C++, Java/JavaFX, SQL, Git & GitHub, Raven, HTML/CSS, FXML, LFDCS, Echoview, MATLAB

CITI & USC Certifications: Physical Science Responsible Conduct of Research; Social and Behavioral Responsible Conduct of Research; Garnet AI Fluency